

STOR 155.010 – Midterm 1 Practice Problems

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Note. This set is for practice, not a preview of exact exam content.

1 Multiple Choice Questions

1. A survey of “favorite campus dining hall” is conducted by a student who stands outside the busiest dining hall at lunchtime and surveys whoever walks by. Which change would most directly fix the main flaw in this design?
 - (A) Survey twice as many people at the same location and time.
 - (B) Randomly select students from the full student directory, regardless of location.
 - (C) Ask each surveyed student to also nominate a friend to survey.
 - (D) Only survey students during dinner instead of lunch.
 - (E) Round the final percentages to the nearest whole number.
2. A company has 12 regional offices, each with 40 employees, and wants employee opinions on a new benefits package. Below are five proposed procedures, each labeled with a sampling method name. Which procedure is mismatched with its label?
 - (A) Simple random sampling: number all 480 employees and randomly select 60.
 - (B) Stratified sampling: randomly select 5 employees from each of the 12 offices.
 - (C) Cluster sampling: randomly choose 3 of the 12 offices and survey every employee there.
 - (D) Convenience sampling: survey only the office located in the same building as company HQ.
 - (E) Stratified sampling: randomly select 3 offices, then survey all employees in each selected office.
3. A survey of 500 car owners records: (1) car color, (2) number of accidents in the past year, (3) owner’s satisfaction rating (1–5 stars), (4) car’s fuel type (gas/electric/hybrid). A student classifies these variables as follows. Which classification is **incorrect**?
 - (A) Car color – Categorical, Nominal
 - (B) Number of accidents – Numeric, Discrete
 - (C) Satisfaction rating – Categorical, Ordinal
 - (D) Fuel type – Numeric, Discrete
 - (E) Number of accidents – not Categorical

4. A researcher fits a least-squares regression predicting exam score from hours studied. She then swaps the roles, refitting a new line to predict hours studied from exam score using the same data. Which of the following will stay exactly the same between the two fitted models?
- (A) The slope
 - (B) The intercept
 - (C) The correlation coefficient r
 - (D) The residual for any specific student
 - (E) The regression equation itself
5. A university observes that students who use the library's 24-hour study room tend to have higher GPAs than those who don't, and wants to determine whether using the room actually causes better grades. Which single change to the study design would most directly allow a causal conclusion?
- (A) Collect a larger sample of students.
 - (B) Track the same students over four years instead of one semester.
 - (C) Randomly assign students to either use or not use the 24-hour room, rather than letting them choose.
 - (D) Ask students to self-report how much they believe the room helped them.
 - (E) Compare only students with similar high school GPAs.
6. Two research teams both study whether a new sleep app improves next-day focus scores. Team 1 randomly assigns all 100 volunteers directly to app/no-app. Team 2 first separates volunteers into "already good sleepers" and "poor sleepers," then randomly assigns app/no-app within each group. If baseline sleep quality strongly affects focus regardless of the app, which team's design will most likely produce more precise estimates of the app's effect?
- (A) Team 1, because larger randomization pools always reduce variability.
 - (B) Team 2, because blocking on a variable that affects the outcome reduces unexplained variability within each comparison.
 - (C) Both teams' designs are equally precise, since both use random assignment.
 - (D) Team 1, because blocking introduces bias into the treatment groups.
 - (E) Neither design can produce a valid causal estimate without a placebo group.
7. A study finds that neighborhoods with more streetlights have lower crime rates. Someone proposes several follow-up analyses to strengthen a causal claim that streetlights reduce crime. Which of the following proposed follow-ups would **NOT** help address the concern that a confounding variable (such as neighborhood income) is driving the association?
- (A) Statistically control for neighborhood income when comparing crime rates.
 - (B) Compare only neighborhoods with similar income levels to each other.
 - (C) Randomly assign new streetlights to some neighborhoods and not others, then compare crime rates.
 - (D) Round the crime rate data to the nearest whole number before analysis.

- (E) Track the same neighborhoods before and after streetlights are installed, controlling for income trends.
8. A gym's dataset of member ages is bimodal, with peaks around age 20 (students) and age 45 (working professionals), and very few members near age 30. If the gym reports only the overall mean age of its members, what practical problem does this create?
- (A) The mean will be mathematically impossible to calculate for a bimodal distribution.
- (B) The reported mean age (around 32–33) may not resemble a typical member at all, since almost no members are actually that age.
- (C) The mean will always equal the median for bimodal data, hiding no information.
- (D) Bimodal distributions cannot have an IQR, so spread cannot be reported either.
- (E) The standard deviation will be negative, since the data has two peaks.
9. A dataset of 6 delivery times (minutes) is: 18, 20, 21, 23, 24, 26. After review, the value 26 is found to be a typo for 62. What is the new mean of the corrected dataset?
- (A) 22.00
- (B) 25.33
- (C) 28.00
- (D) 30.67
- (E) 62.00
10. A dataset has five-number summary: $\text{Min} = 4$, $Q_1 = 10$, $\text{Median} = 14$, $Q_3 = 18$, $\text{Max} = 22$. If a new observation of 33 were added to this dataset, would it be flagged as a potential outlier in the boxplot and why?
- (A) No, because 33 is less than double the maximum.
- (B) Yes, because $IQR = 8$, the upper fence is $18 + 12 = 30$, and 33 exceeds it.
- (C) No, because only values below Q_1 can ever be classified as outliers.
- (D) Yes, because 33 is greater than the median.
- (E) It cannot be determined without the full original dataset.
11. In a strongly left-skewed distribution of employee tenure (years at a company), most employees have long tenure, with a small group of very new hires pulling the tail toward zero. If an employee is at the 90th percentile for tenure, which statement is most accurate?
- (A) This employee has been at the company longer than 90% of employees, and their raw tenure is likely well above the mean, since the mean is pulled down by the new hires.
- (B) This employee has been there for exactly 90% of the maximum possible tenure.
- (C) This employee's raw tenure must be below the median, since percentiles and medians measure different things.
- (D) In a left-skewed distribution, percentile rank cannot be meaningfully interpreted.
- (E) This employee has less tenure than 90% of employees.

12. A dataset of test scores (out of 50) has a mean of 32 and a standard deviation of 6. A teacher rescales the scores to a 100-point scale using the formula $\text{new score} = 2 \times (\text{old score})$. What are the new mean and new standard deviation?
- (A) Mean = 32, SD = 6
 - (B) Mean = 64, SD = 6
 - (C) Mean = 64, SD = 12
 - (D) Mean = 32, SD = 12
 - (E) Mean = 66, SD = 12
13. A mosaic plot displays the relationship between college major (STEM/Non-STEM) and preferred study location (Library/Dorm/Cafe) for 400 students. The STEM column is visibly narrower than the Non-STEM column, and the two columns show different height-splits across study locations. Which conclusion can be defensibly drawn from a mosaic plot alone?
- (A) Being a STEM major caused the difference in study location preference.
 - (B) There are fewer STEM majors than Non-STEM majors in this sample, and the two groups appear to differ in their study location preferences.
 - (C) The exact numeric percentage of STEM majors who prefer the library can be read precisely to the decimal from the plot.
 - (D) A mosaic plot proves independence between major and study location.
 - (E) Mosaic plots cannot be used for a categorical variable with more than two levels.
14. Study A finds a correlation of $r = 0.25$ between two variables using a sample of 40 people. Study B finds a correlation of $r = 0.25$ between two different variables using a sample of 4,000 people. Which statement about the strength of the linear association itself (not the reliability of the estimate) is correct?
- (A) Study B shows a stronger linear association, because of its larger sample size.
 - (B) Both studies show the same strength of linear association, since r measures strength independent of sample size.
 - (C) Study A shows a stronger association, because smaller samples tend to produce more extreme correlations.
 - (D) Correlation cannot be compared across studies with different sample sizes.
 - (E) Study B's r value should be divided by 100 to make it comparable to Study A's.
15. Consider a dataset of paired (X, Y) values. How many of the following four actions would leave the correlation coefficient r completely unchanged?
- I. Adding 100 to every X value.
 - II. Multiplying every Y value by -3 .
 - III. Removing the single observation with the largest X value.
 - IV. Recording X and Y in reverse order (swapping which column is X and which is Y).
- (A) 0
 - (B) 1
 - (C) 2

- (D) 3
- (E) 4
16. A dataset has exactly 3 points: $(2, 10)$, $(4, 6)$, $(6, 2)$. A student claims the correlation between X and Y for this data is exactly $r = -1$. Is the student correct, and how could you verify it quickly without using the full correlation formula?
- (A) No, since r can never equal exactly -1 for real data.
- (B) Yes – check that the three points fall exactly on a single straight line with negative slope, which guarantees $r = -1$.
- (C) No, because there are only 3 points, and r requires at least 5 observations to compute.
- (D) Yes, but only because the X values happen to be evenly spaced.
- (E) It cannot be determined without computing s_x and s_y first.
17. A least-squares regression line predicting Y from X is given as $\hat{y} = 10 + 2.5x$, fitted using data with $\bar{x} = 8$, $s_x = 3$, and $r = 0.6$. What must s_y be?
- (A) 4.5
- (B) 7.5
- (C) 12.5
- (D) 15
- (E) 20.83
18. A regression predicts monthly heating bill (\$) from average outdoor temperature ($^{\circ}\text{F}$): $\hat{y} = 180 - 3.5x$. A student interprets this as: "For every 1 degree increase in temperature, the heating bill decreases by \$3.50, and this means colder temperatures directly cause higher bills due to physics." What is the biggest issue with this interpretation?
- (A) The slope's sign is backwards; it should be positive.
- (B) The numerical interpretation of the slope ($-\$3.50$ per degree) is correct, but claiming this proves a physical causal mechanism goes beyond what a regression line alone can establish.
- (C) Regression lines can only be interpreted for temperatures below freezing.
- (D) The intercept, not the slope, describes the change per degree.
- (E) There is nothing wrong with this interpretation.
19. A regression model predicts blood alcohol content (BAC) from number of drinks consumed, fit using data from people who consumed between 1 and 6 drinks. Which of the following predictions uses the model appropriately (i.e., does **NOT** involve extrapolation)?
- (A) Predicting BAC for someone who had 0 drinks.
- (B) Predicting BAC for someone who had 4 drinks.
- (C) Predicting BAC for someone who had 10 drinks.
- (D) Predicting BAC for someone who had 15 drinks.
- (E) Predicting BAC for someone who had 25 drinks.

20. You are told that a regression line has $R^2 = 0.36$. Which additional single piece of information would let you determine the exact value (including sign) of the correlation coefficient r ?
- (A) The sample size.
 - (B) The sign (positive or negative) of the slope of the regression line.
 - (C) The value of the y -intercept.
 - (D) The units of the response variable.
 - (E) The mean of the explanatory variable.
21. A regression line is fit to a dataset, and it is given that the line passes through $(\bar{x}, \bar{y})$ as it always does. If a particular observation happens to have $x_i = \bar{x}$ and $y_i = \bar{y}$ exactly, what is the residual for this observation?
- (A) 0
 - (B) Equal to the slope of the line
 - (C) Equal to the intercept of the line
 - (D) Equal to $\bar{y}$
 - (E) It cannot be determined without knowing the equation of the line
22. A regression is fit to 40 data points, giving a slope of 2.1. After removing one specific data point (which had an extreme x -value far from the rest), the slope is refit using the remaining 39 points and comes out to 2.08. Based on this evidence alone, how should the removed point most likely be classified?
- (A) An influential point, since removing it changed the slope.
 - (B) A high leverage point that was **not** particularly influential, since despite its extreme x -value, removing it barely changed the slope.
 - (C) A residual outlier, since the slope changed at all.
 - (D) This evidence proves the point was a data-entry error.
 - (E) No classification is possible without recomputing R^2 .
23. The table below shows residuals from a regression model at several values of the explanatory variable x :

x	Residual
5	-0.2
10	0.3
15	-1.8
20	4.5
25	-6.1

Which condition for a valid least-squares model does this pattern of residuals most likely violate?

- (A) Linearity
- (B) Constant variability

- (C) Independence of observations
 - (D) The response variable must be positive
 - (E) None of the conditions are violated
24. A data analyst knows their dataset likely contains a few extreme, erroneous y -values that couldn't be corrected before a deadline. They must choose between two regression approaches: one that minimizes the sum of squared residuals, and one that minimizes the sum of absolute residuals. Which approach is the better choice given their situation, and why?
- (A) Minimize squared residuals, because squaring always produces more accurate models.
 - (B) Minimize absolute residuals, because this method is less sensitive to a few extreme y -values than squaring the residuals.
 - (C) Minimize squared residuals, because it is less sensitive to extreme values than the absolute-residual method.
 - (D) Both methods are equally sensitive to extreme values, so it does not matter which is chosen.
 - (E) Neither method should be used if the dataset contains any errors at all.

2 Free Response Questions

1. A tech company reports that employees who used a new AI coding assistant completed 30% more tasks per week than employees who did not, and concludes the assistant *causes* increased productivity.
 - (a) Was this most likely an observational study or an experiment? Explain how you can tell.
 - (b) Identify one confounding variable that could explain the productivity difference without the AI assistant being the cause.
 - (c) Briefly describe how you would redesign this as a randomized experiment to better support a causal claim.
2. A campus dining survey asks students their preferred meal type (Breakfast, Lunch, Dinner, Late-night) and separately records each student's dorm building and which dining hall they visit most often.
 - (a) Would a bar plot or a pie chart make it easier to precisely compare the two least popular meal types? Explain.
 - (b) Would a contingency table or a single histogram be the appropriate way to summarize the relationship between dorm building and preferred dining hall? Explain why the other option would not work.
3. A podcast platform records episode length (minutes) for 300 episodes. Most episodes run 20–45 minutes, but a small number of long-form interview specials run over 2 hours, pulling the distribution's tail to the right. Would you recommend the mean or the median to describe a "typical" episode length? Would you recommend the standard deviation or the IQR to describe the spread? Explain your reasoning, and state whether you'd expect the mean to be greater than, less than, or about equal to the median.
4. Daily commute times (minutes) for a group of employees have $Q_1 = 15$, median = 22, and $Q_3 = 30$.
 - (a) Calculate the IQR and the upper and lower whiskers for flagging potential outliers.
 - (b) One employee's commute took 55 minutes due to a highway closure. Would this be flagged as a potential outlier? Show your work.
5. A university's fitness center posts a QR code survey only inside the gym, asking respondents how many days per week they exercise. Explain what type of bias this introduces, and describe the direction you'd expect it to push the reported average number of exercise days, relative to the true average for the whole student body.
6. A fitness tracker company studies the relationship between average daily step count in thousands (x) and a sleep quality score out of 100 (y) for 150 users. Summary statistics: $\bar{x} = 8$, $s_x = 3$, $\bar{y} = 70$, $s_y = 12$, $r = 0.5$.
 - (a) Interpret $r = 0.5$ in context.
 - (b) Find the least-squares regression line predicting sleep quality from step count, and interpret the slope.
 - (c) Calculate and interpret R^2 .
 - (d) One user averaged 10,000 steps/day and had an actual sleep quality score of 95. Calculate this user's residual, and state whether the model over- or under-predicted their score.

- (e) A different user averaged 20,000 steps/day (training for a marathon) and had a sleep quality score of only 40, far below what the model predicts, due to muscle soreness. Classify this data point as a residual outlier, high leverage point, or influential point, and justify your answer.
7. A streaming service fits a regression line predicting monthly subscriber churn rate (%) from subscription price (\$), using prices ranging from \$5 to \$20. A residual plot shows residuals tightly clustered near zero for low prices, but much more spread out (both positive and negative) for higher prices. Which condition required for a valid least-squares model does this violate? What does this imply about the reliability of predictions at higher price points?
8. Five students report hours studied for a quiz (x) and their quiz score out of 100 (y):

Hours studied (x)	Quiz score (y)
1	50
2	60
3	65
4	80
5	95

- (a) Calculate the sample mean and sample median of x and of y .
- (b) Find the least-squares regression line $\hat{y} = b_0 + b_1x$. Show all steps.
- (c) Using your regression line, calculate the predicted quiz score for the student who studied 3 hours. Then calculate and interpret the residual for this student.
9. An analyst studies electric vehicle battery size (kWh) and driving range (miles) across 25 models, then fits a regression line. One ultra-luxury model has a battery size far larger than any other car in the dataset, but its actual range falls almost exactly where the regression line already predicted.
- (a) Classify this vehicle as a residual outlier, high leverage point, or influential point. Justify your answer.
- (b) Suppose instead this same vehicle's actual range were far below what the line predicts. Would your classification change? Explain.
10. A histogram of daily step counts for 5,000 fitness app users shows two distinct peaks: one around 4,000 steps and another around 12,000 steps. What term describes this shape? Propose an explanation, in terms of possible subgroups within the users, for why the distribution might look this way.